Sum and difference of cubes

1 If $a^3 + 125 = (a+5)\left(a^2 + ka + 25\right)$, find the value of k.

2 Factor:

a $x^3 + y^3$ **b** $x^3 - y^3$ **c** $x^3 + 1$ **d** $x^3 - 8$

e $x^3 + 27$ **f** $x^3 - 64$ **g** $x^3 + 512$ **h** $x^3 - 1000$

3 Factor:

a $x^3 + 64y^3$ **b** $x^3 - 125y^3$ **c** $x^3 + 729y^3$ **d** $x^3 - 216y^3$

e $x^3 + 1000y^3$ **f** $a^3 + 27b^3$ **g** $s^3 + 125t^3$ **h** $u^3 - 343v^3$

4 Factor:

a $27x^3 + 64$ **b** $27x^3 - 8$

c $1000x^3 + 729$ **d** $512x^3 - 343$

e $8x^3 + 1000$ **f** $125x^3 - 1$

g $512x^3 + 8$ **h** $216x^3 - 64$

5 Factor:

a $250u^3 + 54v^3$ **b** $54m^3 - 250n^3$

c $27x^3 + 8y^3$ **d** $125x^3 - 64y^3$

e $216x^3 + 343y^3$ **f** $343x^3 - 1000y^3$

g $40u^3 + 5v^3$ **h** $4m^3 - 32n^3$

6 Factor:

a $(x+7)^3 + (x-8)^3$ **b** $(x-3)^3 - (x+2)^3$

c $(x-2)^3 + (x+5)^3$ **d** $(x+1)^3 - (x-7)^3$

7 Simplify the following:

a $\dfrac{125x^3 + 8}{5x + 2}$ **b** $\dfrac{x^3 - 125}{4x - 20}$

c $\dfrac{x^3 - 125}{x^2 - 25}$ **d** $\dfrac{8x^3 - 27y^3}{2x - 3y}$

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8 Consider the polynomial $x^6 - y^6$.

a By thinking of the polynomial as $\left(x^3\right)^2 - \left(y^3\right)^2$, fully factor it starting with a difference of two squares.

b By thinking of the polynomial as $\left(x^2\right)^3 - \left(y^2\right)^3$, show that we can use a difference of two cubes to write it in the form $(x-y)(x+y)\left(x^4 + x^2y^2 + y^4\right)$.

c By comparing the results of parts (a) and (b), write down a factoring for $x^4 + x^2y^2 + y^4$.

9 Evaluate $11^3 + 6^3$ by factorising it using the sum of two cubes method.

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